

Deepak Mallampati

FORWARD DEPLOYED / APPLIED AI ENGINEER

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |
US-based, F-1 OPT (open to relocation/sponsorship)

PROFESSIONAL SUMMARY

Applied AI engineer with 4+ years shipping production LLM and agentic systems and integrating them into real products across regulated enterprises. Builds and deploys AI features alongside product, platform, and data teams - owning REST API design, integration, and adoption end-to-end (12 APIs powering 100,000+ requests/day across 20+ teams). Comfortable embedding with stakeholders as the trusted technical expert, translating novel AI architectures into working systems quickly. US-based on F-1 OPT; open to relocation/sponsorship for the right Forward Deployed role.

CORE COMPETENCIES

Forward Deployed / Solutions Engineering • REST API Development & Integration • Agentic AI & LLM Systems • Customer / Stakeholder-Facing Delivery • RAG & Retrieval • FastAPI / Python / JavaScript • Cloud (AWS, Azure, GCP) • Observability & Eval

Languages & APIs: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | REST API design & integration

Generative AI & LLMs: GPT-4o/5, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | RAG | Multi-Agent Systems | Prompt Engineering

Platform & MLOps: AWS (Bedrock, SageMaker, Lambda), Azure OpenAI, GCP Vertex | Docker, Kubernetes | CI/CD | MLflow, W&B | Monitoring

Retrieval & Data: FAISS, Pinecone, Weaviate | Cross-encoder re-ranking | PostgreSQL, MongoDB, Redis | Kafka, Spark

Security & Compliance: HIPAA, SOC 2, PCI DSS | OAuth 2.0, JWT, RBAC | PII redaction | Prompt-injection mitigation

PROFESSIONAL EXPERIENCE

AI Engineer - Progress Solutions

Jul 2025 - Present | USA

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents - reducing token consumption by **42%** while sustaining accuracy across multi-step workflows.
- Engineered high-performance **RAG pipelines** over large healthcare corpora (dense retrieval + cross-encoder re-ranking) to lift relevance and materially reduce hallucination on domain queries.
- Built fault-tolerant **automation infrastructure** with scheduled orchestration and exponential-backoff retry, reducing failed production runs by **60%**.
- Established an **LLM evaluation & monitoring framework** (RAGAS, BERTScore, custom metrics) with latency/accuracy dashboards to catch regressions before release.

AI Engineer Intern - Vanguard

Jan 2024 - Jan 2025 | USA

- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by **40%**.
- Partnered with product, platform, and data engineering teams to deliver scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams - gathering requirements, building integrations, and driving adoption.
- Developed and enhanced **AI agents and backend services** for Vanguard's internal AI platform across **25+ agents** and workflows.
- Engineered safeguards and content controls for secure, policy-compliant AI responses, reducing unsafe outputs by **42%** while maintaining quality.

ML & Gen AI Research Assistant - Kent State University

Oct 2023 - Jan 2024 | USA

- Built a privacy-preserving ML pipeline (k-anonymity, l-diversity, differential privacy), cutting re-identification risk from **3.38% to 0.32%** while retaining **99%** of baseline predictive performance.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an H100 for controllable long-form generation; authored an IEEE-format paper.

ML Trainee - Emerson

Jun 2022 - Apr 2023 | India

- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification on domain-specific data, achieving **89% F1** on custom NER.
- Built regression/classification models for equipment-failure prediction and anomaly detection on operational data.

PROJECTS

MangaLens

Shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation across **29+ sites** - a real product with real users, owned end-to-end.

career-ops

Autonomous AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 | GPA 3.70/4.0